

Erdős Problem #1074

EP-1074

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let S be the set of all $m \geq 1$ such that there exists a prime $p \equiv 1 \pmod{m}$ such that $m! + 1 \equiv 0 \pmod{p}$. Does $\lim_{x \rightarrow \infty} \frac{|S \cap [1, x]|}{x}$ exist? What is it?

Similarly, if P is the set of all primes p such that there exists an m with $p \equiv 1 \pmod{m}$ such that $m! + 1 \equiv 0 \pmod{p}$, then does $\lim_{x \rightarrow \infty} \frac{|P \cap [1, x]|}{\pi(x)}$ exist? What is it?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>